Lab Manual 3

1. University Management System

a. Entities & Attributes

University → University_ID, Name

Department → Department_ID, Name

Student → Student_ID, Name, Email, Phone

Faculty → Faculty_ID, Name, Designation

Course → Course_ID, Course_Name, Credits

Enrollment (associative) → Enrollment_ID, Student_ID, Course_ID

b. Primary Keys

University → University_ID

Department → Department_ID

Student → Student_ID

Faculty → Faculty_ID

Course → Course_ID

Enrollment → Enrollment_ID

c. Relationships & Cardinality

University → Department → 1 : M

Department → Course → 1 : M

Department → Student → 1 : M

Faculty → Course → 1 : M

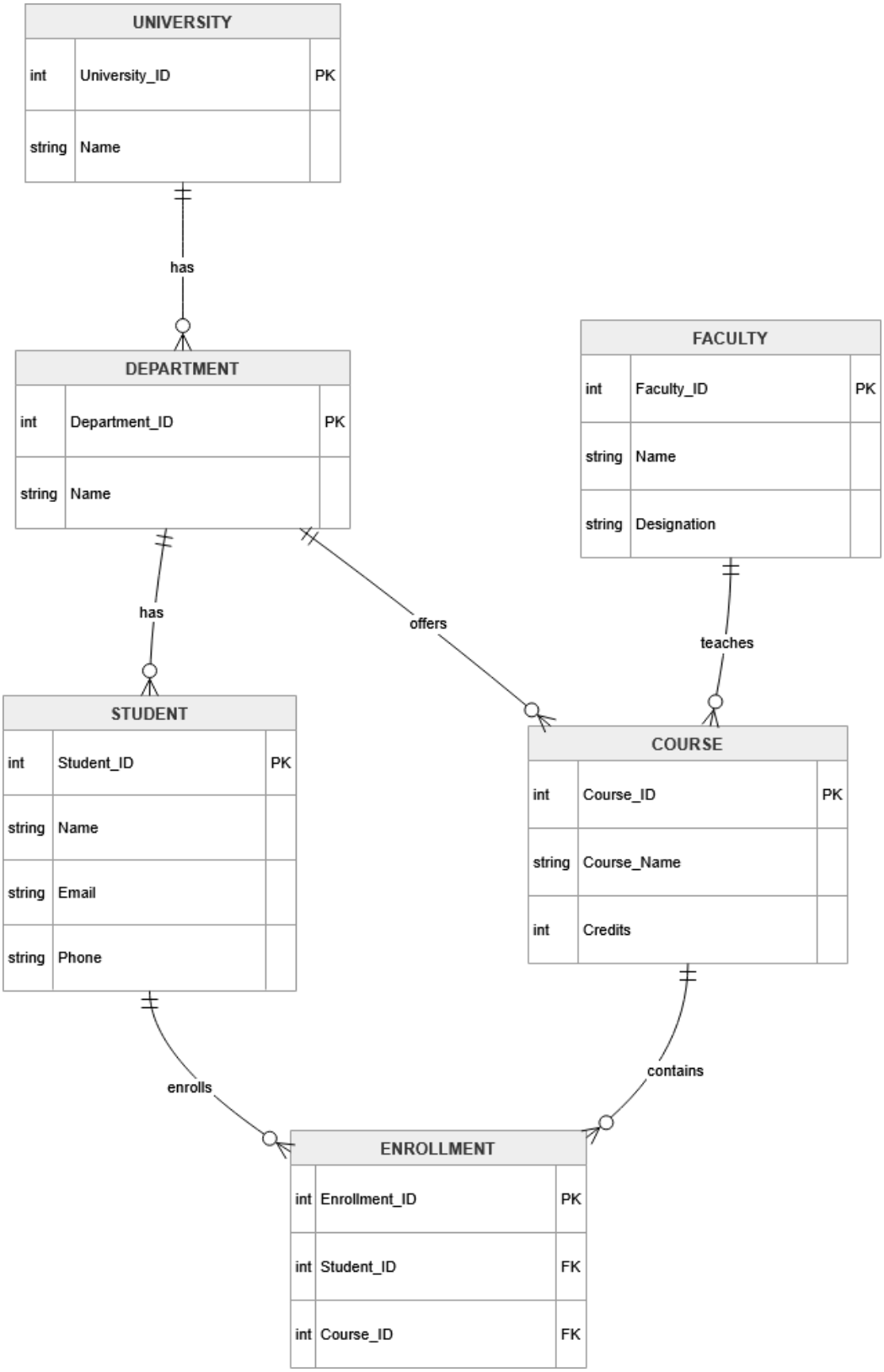
Student → Course → M : N (via Enrollment)

d. M:N Resolution

Student ↔ Course → resolved using Enrollment

Contains FK: Student_ID, Course_ID

e. ERD



2. Online Shopping System

a. Strong and Weak Entities

Strong Entities:

Customer, Order, Product, Category, Payment

Weak / Associative Entity:

Order_Item (depends on Order + Product)

b. Relationships & Cardinality

Customer → Order → 1 : M

Order → Product → M : N (via Order_Item)

Order → Payment → 1 : 1

Category → Product → 1 : M

c. M:N Resolution

Order ↔ Product → resolved using Order_Item

Attributes: Order_ID, Product_ID, Quantity

d. Participation Constraints

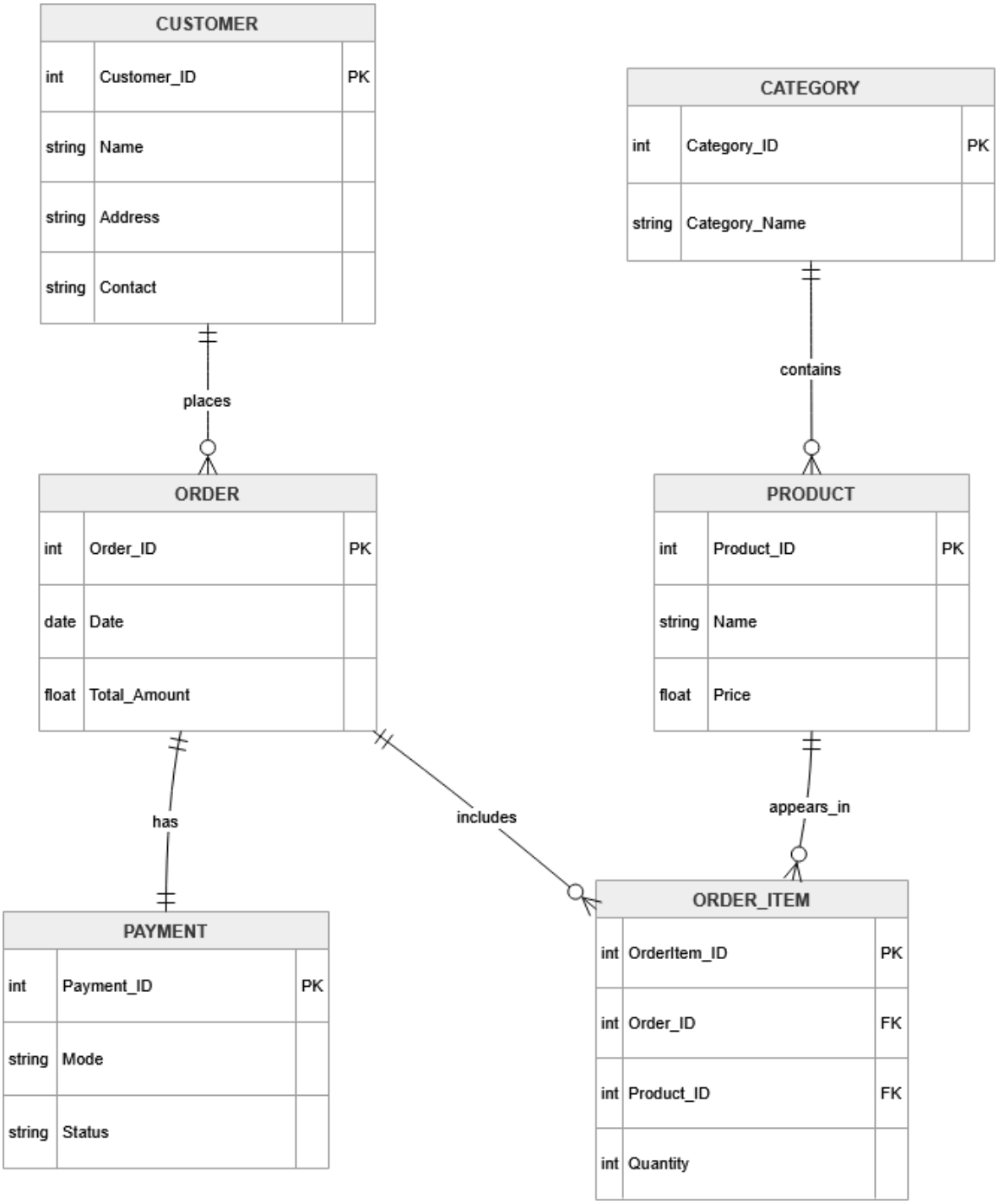
Customer → Order → Partial (customer may not order)

Order → Payment → Total (every order must have payment)

Order → Order_Item → Total

Product → Order_Item → Partial

e. ERD


CUSTOMER
int Customer_ID PK
string Name
string Address
string Contact
places
CATEGORY
int Category_ID PK
string Category_Name
contains
ORDER
int Order_ID PK
date Date
float Total_Amount
PRODUCT
int Product_ID PK
string Name
float Price
has
includes
appears_in
PAYMENT
int Payment_ID PK
string Mode
string Status
ORDER_ITEM
int OrderItem_ID PK
int Order_ID FK
int Product_ID FK
int Quantity